

1

2

3

4

A

A

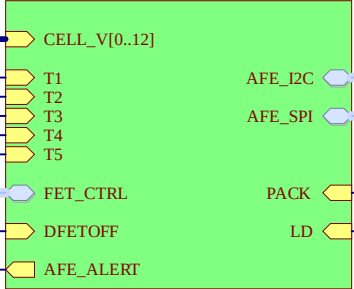
Battery Interface
Battery_Interface.SchDoc



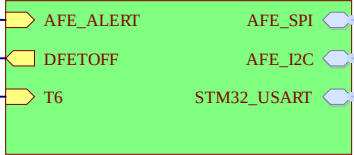
Temperature Sense
Temperature_Sense.SchDoc



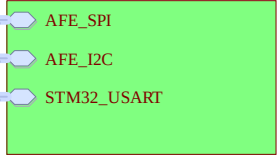
AFE
BQ76952_AFE.SchDoc



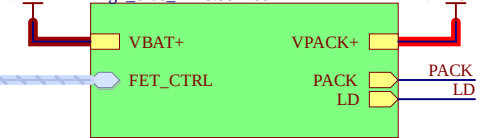
STM32 MCU
STM32_MCU.SchDoc



Pack Interface
Pack_Interface.SchDoc



High Side FETs
High_Side_FETs.SchDoc



Mounting Holes

MH1

MH2

MH3

MH4

1

2

3

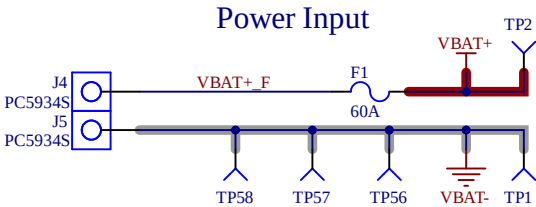
4

D

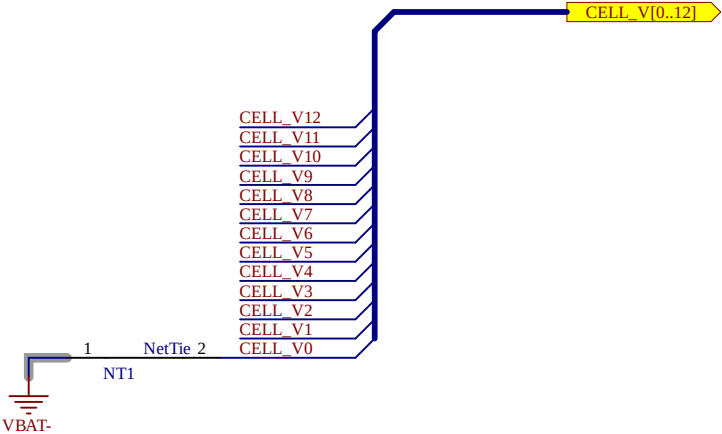
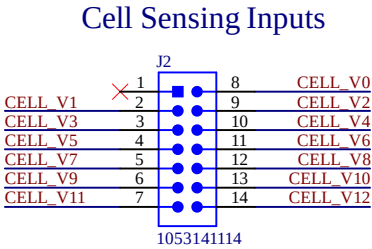
D

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PROJECT LTO 12S BMS.PrjPcb, [No Variations]		
DOCUMENT Top.SchDoc		MODIFIED 2026-05-18
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BATTERY INTERFACE

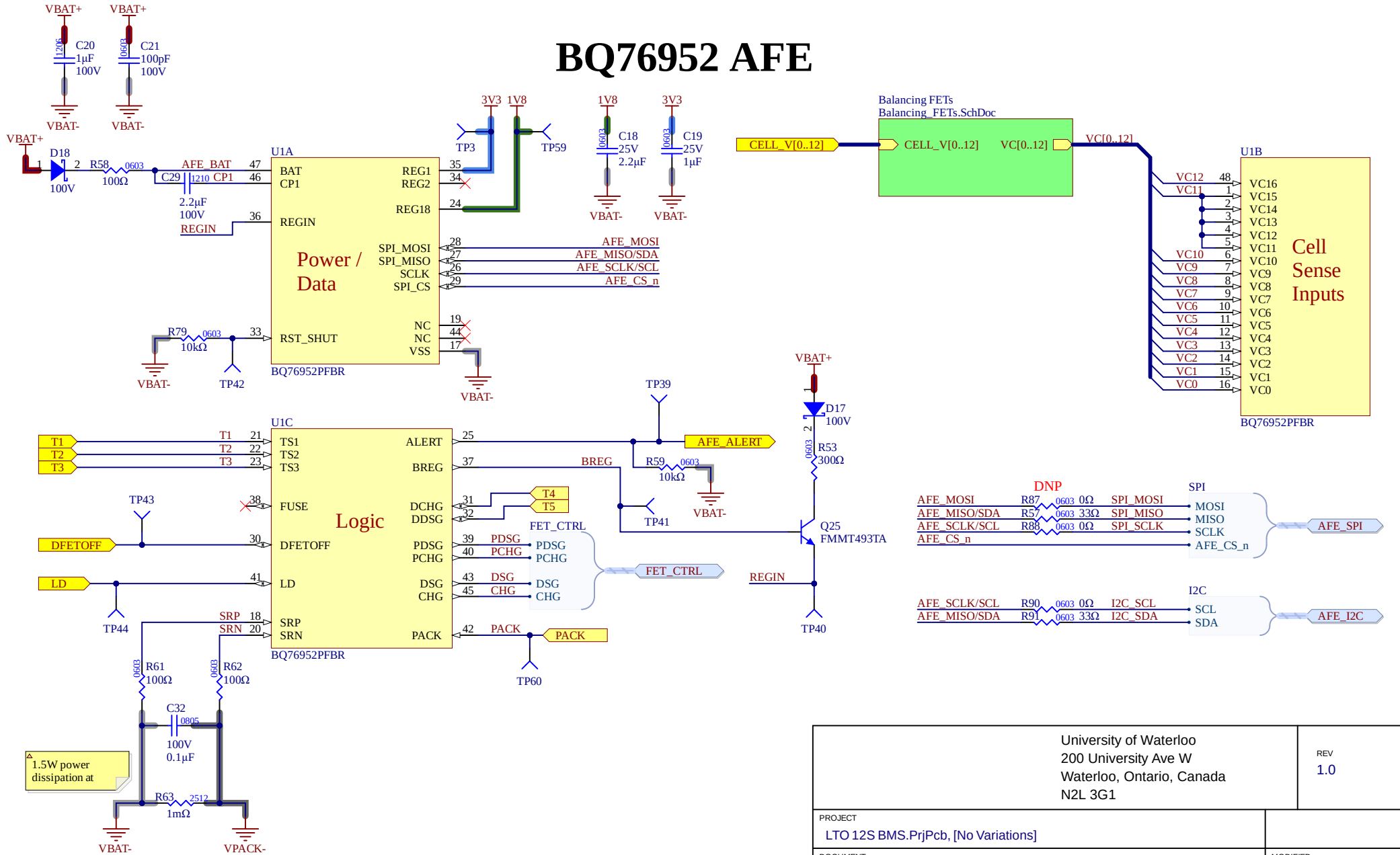


Battery specs:
- 12s1p 20Ah LTO battery
- 1.5V - 2.7V cell voltage range, 2.3V nominal



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BQ76952 AFE

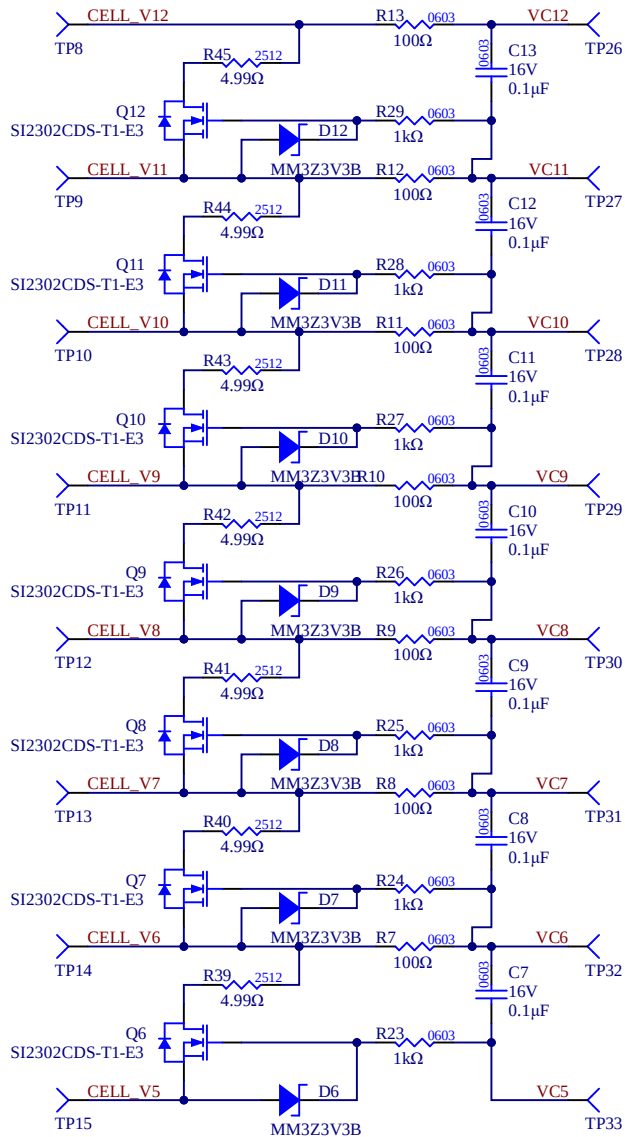


PROJECT LTO 12S BMS.PrjPcb, [No Variations]		REVIEWER *	
DOCUMENT BQ76952_AFE.SchDoc		MODIFIED 2026-05-18	
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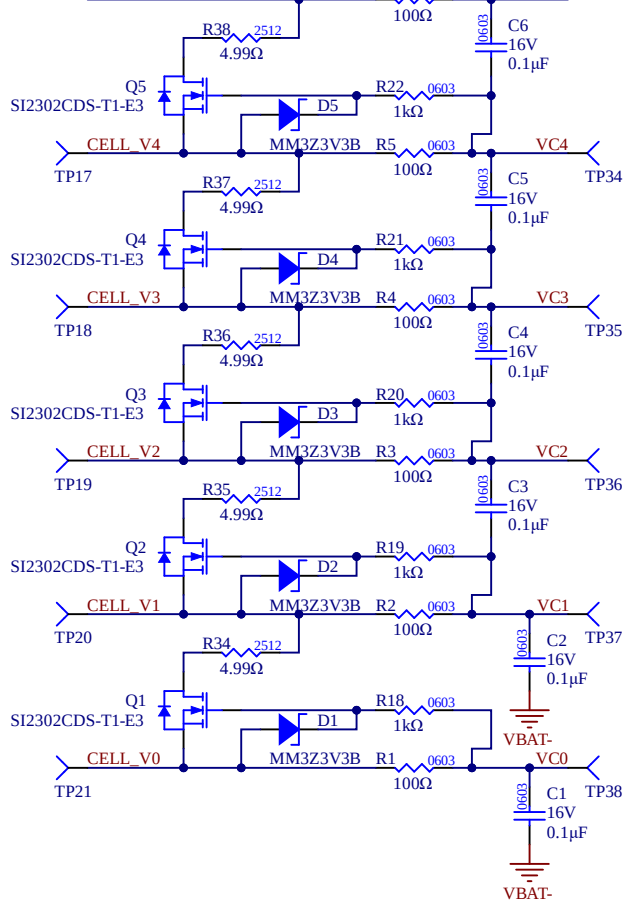
BALANCING FETS

CELL_V[0..12]

- CELL_V12
- CELL_V11
- CELL_V10
- CELL_V9
- CELL_V8
- CELL_V7
- CELL_V6
- CELL_V5
- CELL_V4
- CELL_V3
- CELL_V2
- CELL_V1
- CELL_V0



CELL_V5

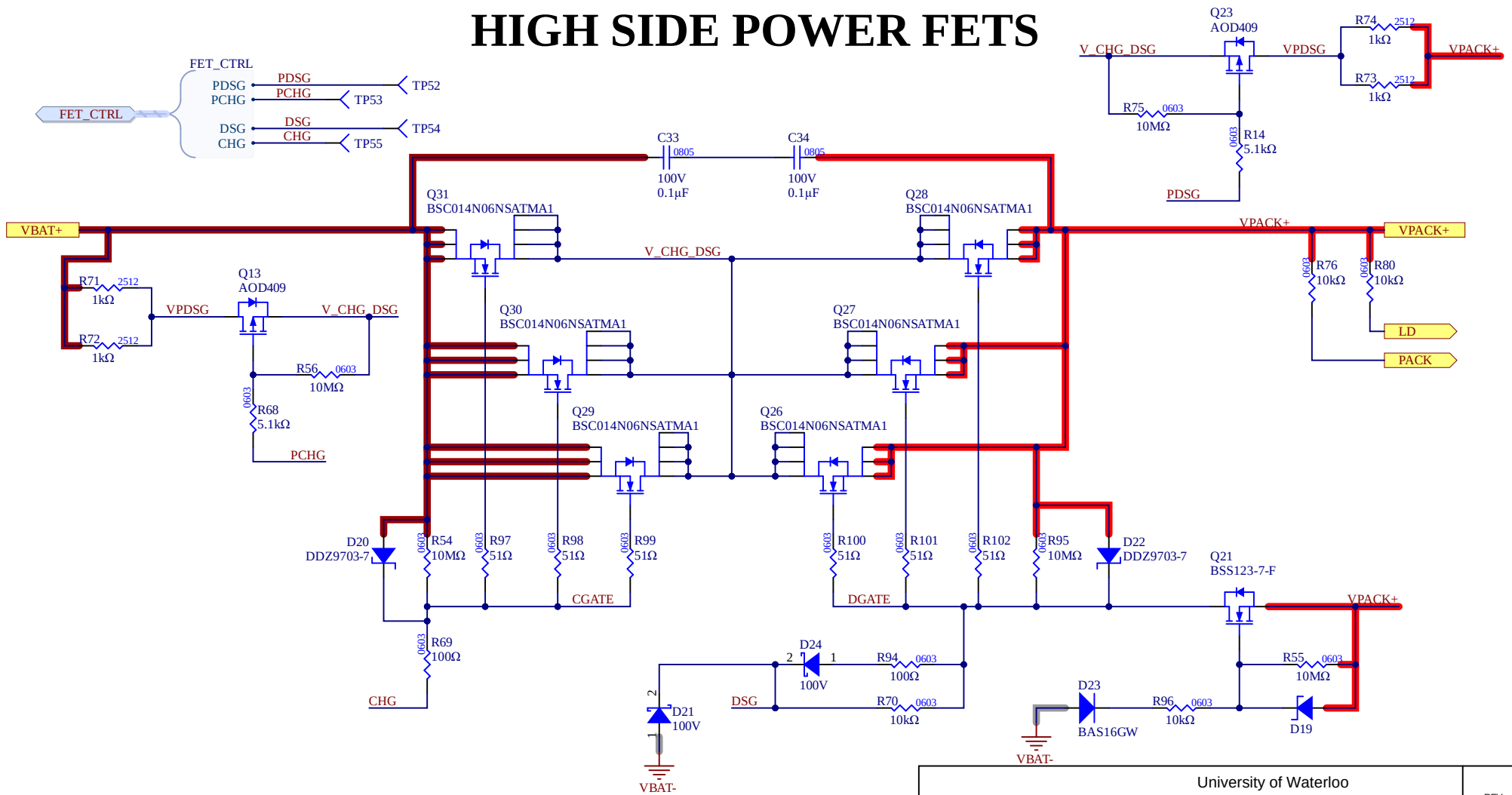


- VC12
- VC11
- VC10
- VC9
- VC8
- VC7
- VC6
- VC5
- VC4
- VC3
- VC2
- VC1
- VC0

VC[0..12]

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HIGH SIDE POWER FETS



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1.0

PROJECT	LTO 12S BMS.PrjPcb, [No Variations]
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DOCUMENT
High_Side_FETs.SchDoc

MODIFIED	2026-05-18
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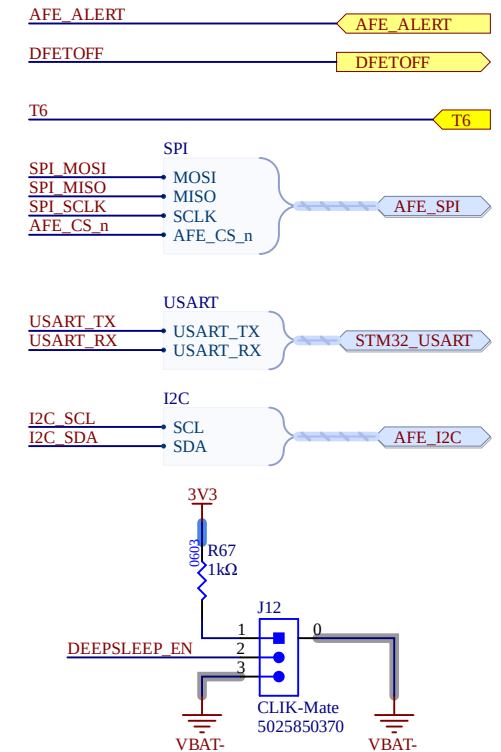
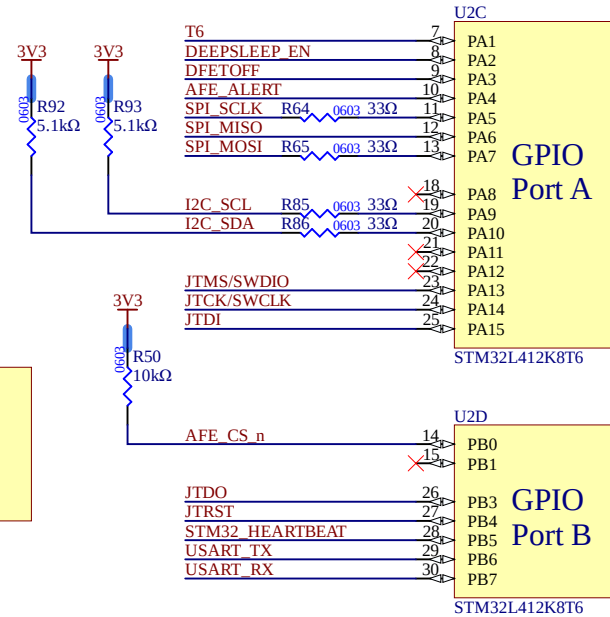
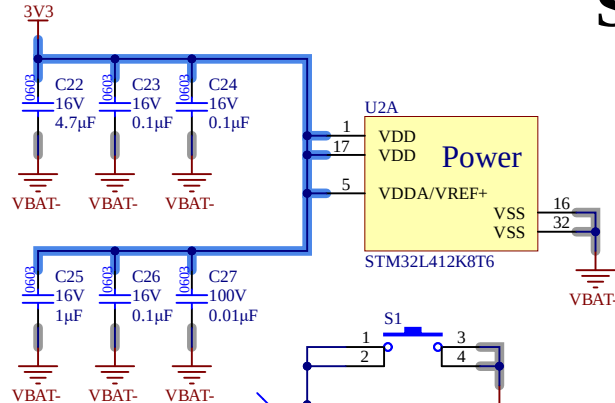
ENGINEER
Farris Matar

	REVIEWER
	★

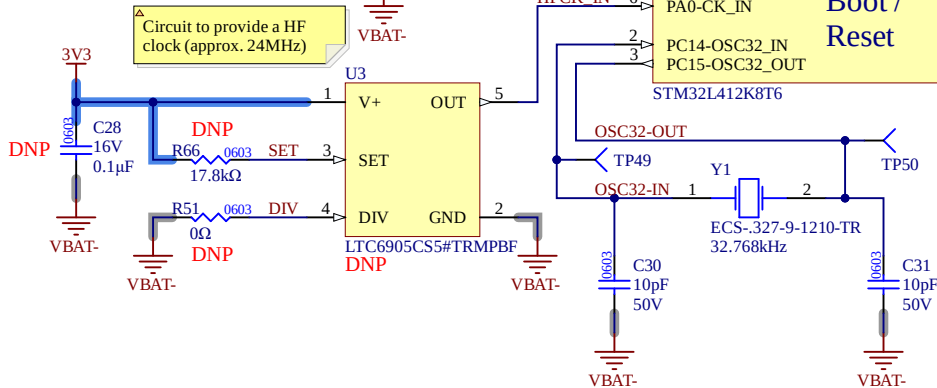
SHEET 5 OF 8

STM32 MCU

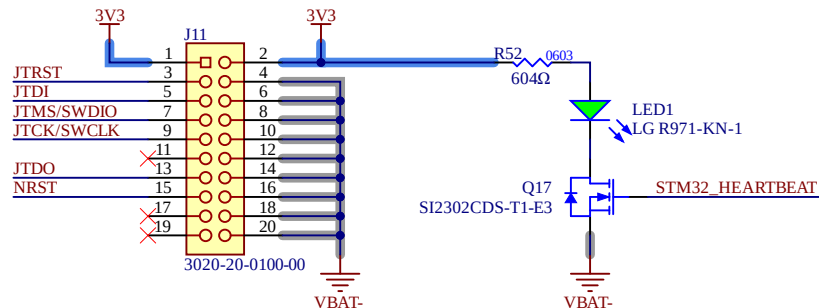
Follows reference design provided in
https://www.st.com/resource/en/application_note/an4555-getting-started-with-stm32l4-series-and-stm32l4-series-hard-ware-development-stmicroelectronics.pdf (Figure 14), minor modifications made to fit selected package



Connector to connect to a toggle switch that will trigger STM32 to put AFE in DEEPSLEEP (very low current consumption) and also put STM32 in SHUTDOWN mode until switch is flipped to wake it back up

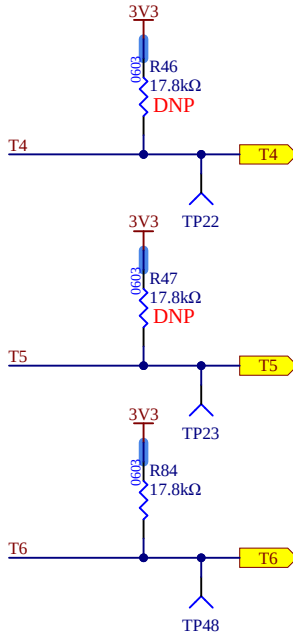
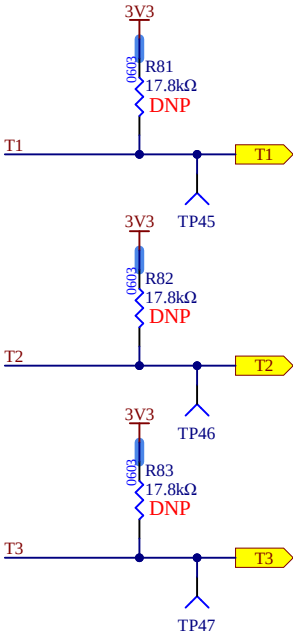
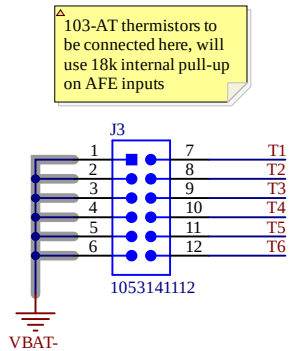


Debug / Programming Connector



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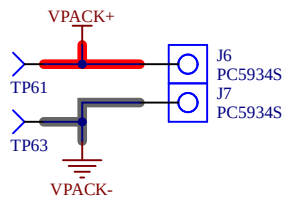
TEMPERATURE SENSING



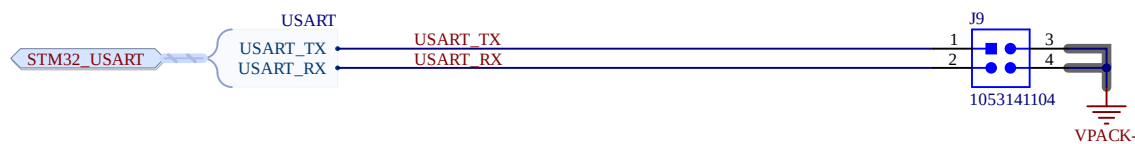
University of Waterloo 200 University Ave W Waterloo, Ontario, Canada N2L 3G1		REV 1.0
PROJECT LTO 12S BMS.PrjPcb, [No Variations]		
DOCUMENT Temperature_Sense.SchDoc		MODIFIED 2026-05-18
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PACK INTERFACE

Power Interface



Data Output



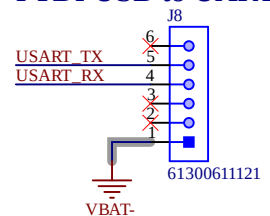
▲ UART will be used to interface with the main control board, sending battery info & state and receiving any commands from the main control board/user

SPI Debug



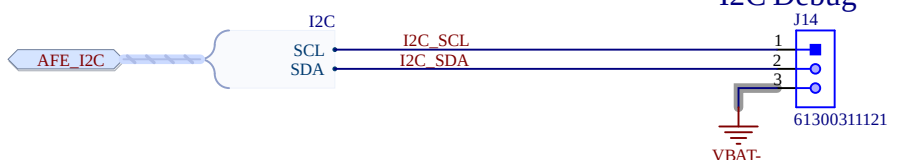
DNP

FTDI USB to UART Debug



▲ Breakout headers for debugging/testing

I2C Debug



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DOCUMENT Pack_Interface.SchDoc		MODIFIED 2026-05-18
ENGINEER Farris Matar	REVIEWER *	SHEET 8 OF 7